

Permutation

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Dependencies

This document depends on the following packages:

```
library(devtools)
library(Biobase)
library(limma)
library(edge)
library(genefilter)
library(qvalue)
```

To install these packages you can use the code (or if you are compiling the document, remove the `eval=FALSE` from the chunk.)

```
install.packages(c("devtools"))
source("http://www.bioconductor.org/biocLite.R")
biocLite(c("Biobase", "limma", "genefilter", "jdstorey/edge", "qvalue"))
```

Download the data

Here we are going to use some data from the paper Evaluating gene expression in C57BL/6J and DBA/2J mouse striatum using RNA-Seq and microarrays. (<http://www.ncbi.nlm.nih.gov/pubmed?term=21455293>) that is a comparative RNA-seq analysis of different mouse strains.

```
con =url("http://bowtie-bio.sourceforge.net/recount/ExpressionSets/bottomly_eset.RData")
load(file=con)
close(con)
bot = bottomly.eset
pdata=pData(bot)
edata=as.matrix(exprs(bot))
fdata = fData(bot)
ls()
```

```
## [1] "bot"          "bottomly.eset" "con"          "edata"
## [5] "fdata"        "pdata"         "tropical"
```

Transform the data

Here we will transform the data and remove lowly expressed genes.

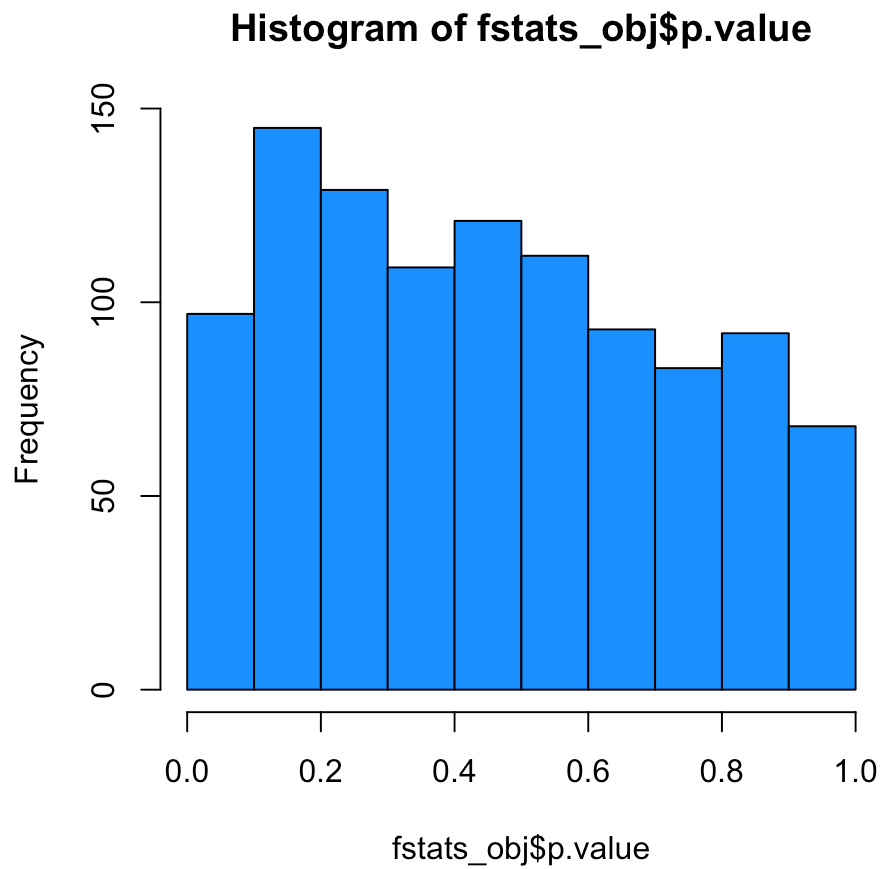
```
edata = log2(as.matrix(edata) + 1)
edata = edata[rowMeans(edata) > 10, ]
```

Calculate p-values parametrically

There are a number of ways to calculate p-values directly. Here are a couple of examples, but there are many data-specific ways of calculating them using packages like `snpStats` or `DESeq2` or `diffbind`.

With genefilter

```
fstats_obj = rowFtests(edata,as.factor(pdata$strain))
hist(fstats_obj$p.value,col=2)
```

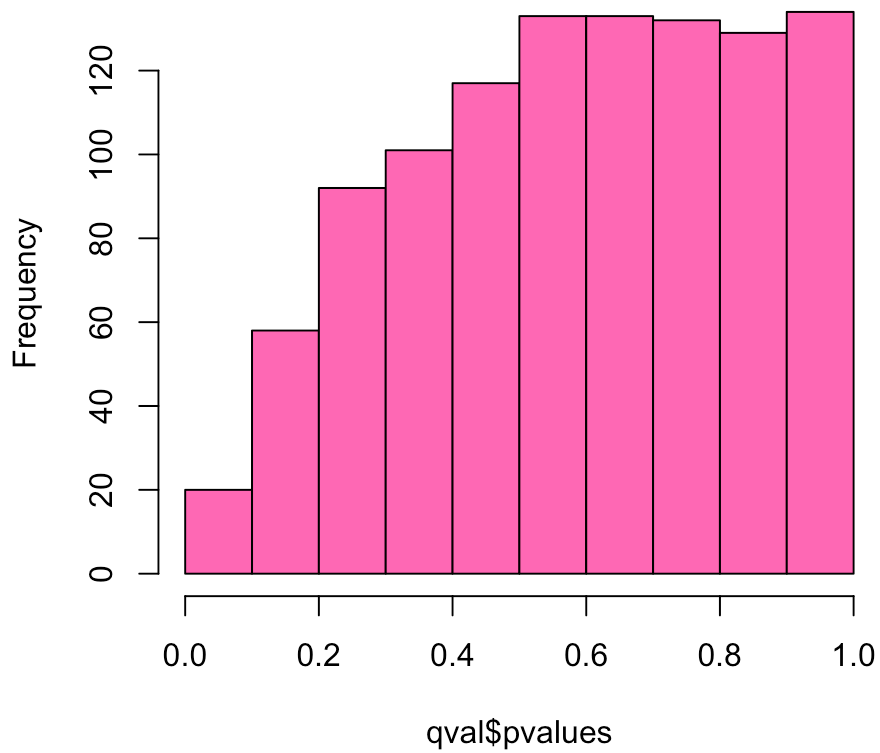


Adjusting for variables with edge

If you want to adjust for variables you need to use `edge`

```
edge_study = build_study(edata, grp = pdata$strain,  
                          adj.var = as.factor(pdata$lane.number))  
de_obj = lrt(edge_study)  
qval = qvalueObj(de_obj)  
hist(qval$pvalues,col=3)
```

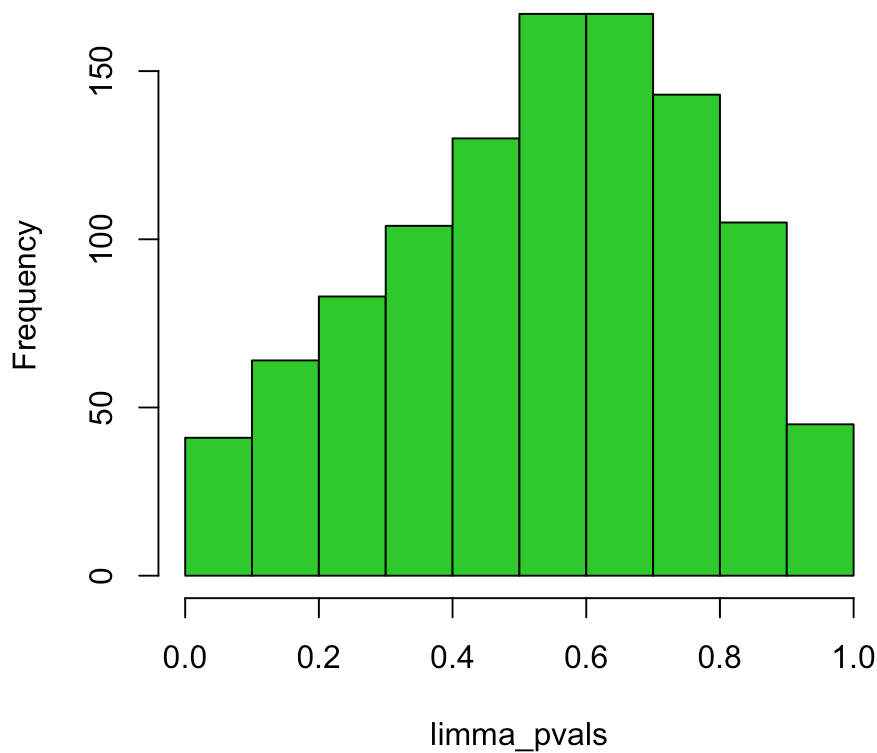
Histogram of qual\$pvalues



P-values for moderated statistics with limma

```
mod = model.matrix(~ pdata$strain + pdata$lane.number)
fit_limma = lmFit(edata,mod)
ebayes_limma = eBayes(fit_limma)
limma_pvals = topTable(ebayes_limma,number=dim(edata)[1])$P.Value
hist(limma_pvals,col=4)
```

Histogram of limma_pvals



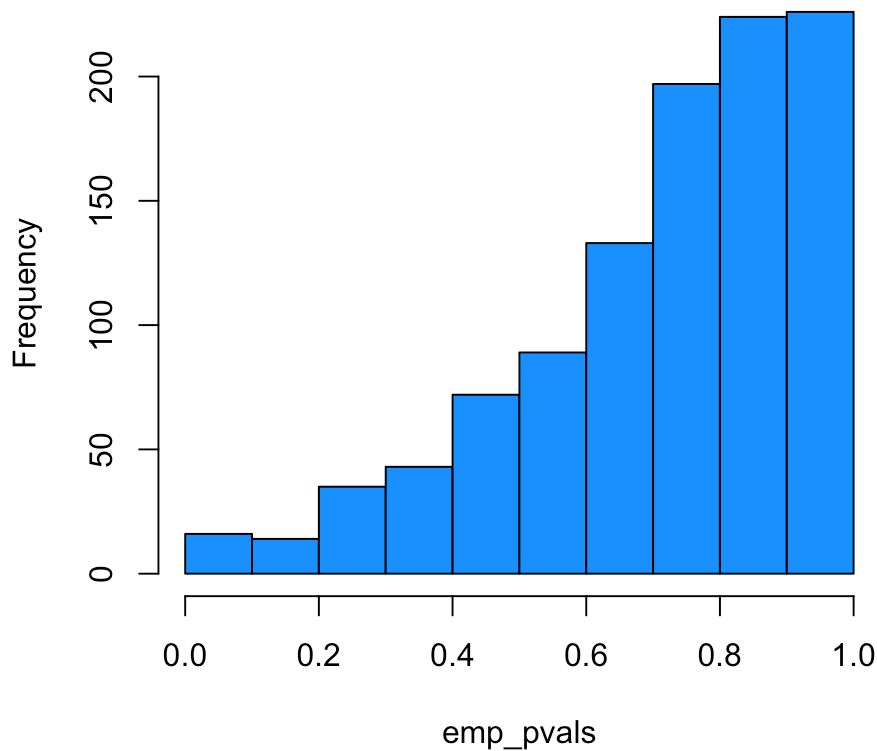
Calculating empirical permutation p-values with edge

Often when you permute you are trying to calculate an empirical p-value. To do this we can compare each observed statistic to the permuted statistics. You can either compare within a single gene (argument `pooled=FALSE` in the `empPvals` function) or pooling the permuted statistics across multiple genes (argument `pooled=TRUE` in the `empPvals` function, the default).

```
set.seed(3333)
B = 1000
tstats_obj = rowttests(edata,pdata$strain)
tstat0 = matrix(NA,nrow=dim(edata)[1],ncol=B)
tstat = tstats_obj$statistic
strain = pdata$strain
for(i in 1:B){
  strain0 = sample(strain)
  tstat0[,i] = rowttests(edata,strain0)$statistic
}

emp_pvals = empPvals(tstat,tstat0)
hist(emp_pvals,col=2)
```

Histogram of emp_pvals



Multiple testing

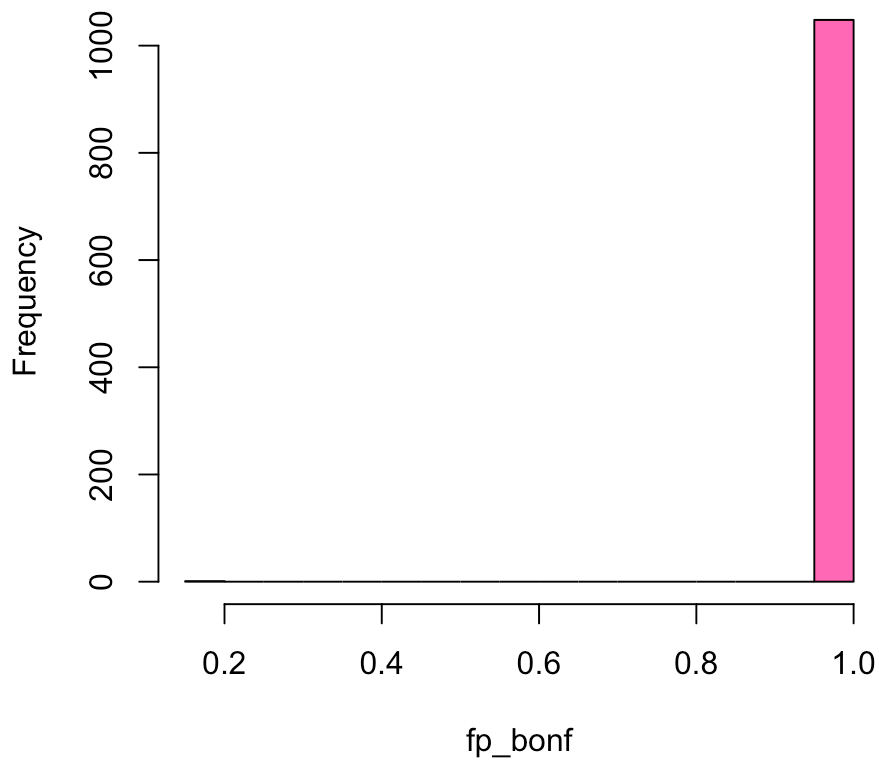
To correct for multiple testing you can use the Bonferroni correction or different FDR corrections.

Bonferroni and Benjamini-Hochberg FDR correction with p.adjust

You can use the `p.adjust` function to get “multiple testing corrected” p-values which you can then use to control error rates.

```
fp_bonf = p.adjust(fstats_obj$p.value,method="bonferroni")
hist(fp_bonf,col=3)
```

Histogram of fp_bonf

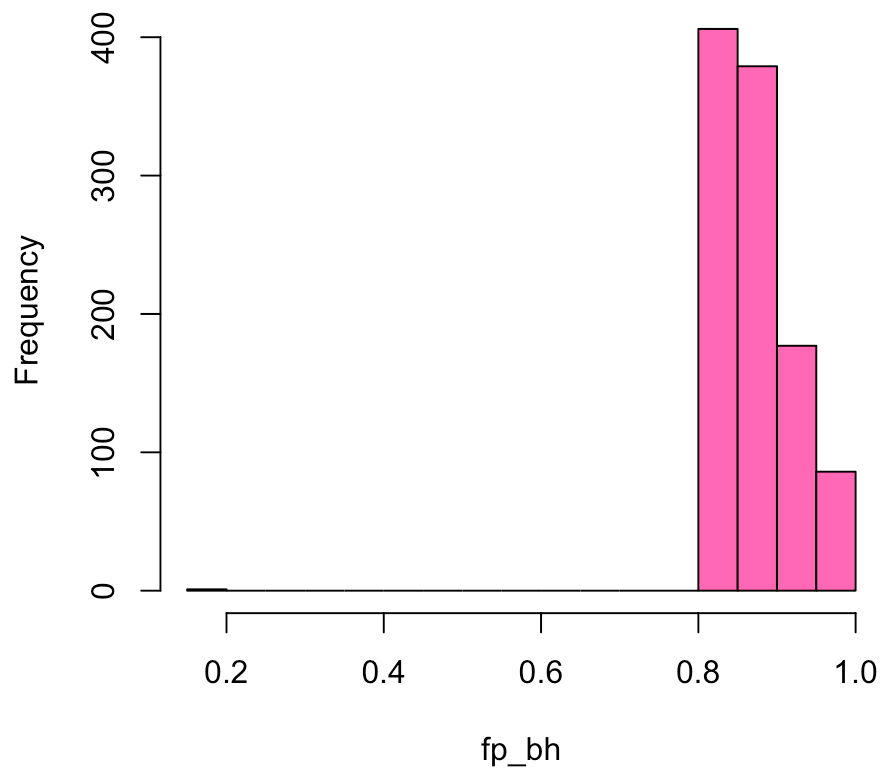


```
quantile(fp_bonf)
```

```
##           0%          25%          50%          75%          100%  
## 0.1791152 1.0000000 1.0000000 1.0000000 1.0000000
```

```
fp_bh = p.adjust(fstats_obj$p.value,method="BH")  
hist(fp_bh,col=3)
```

Histogram of fp_bh



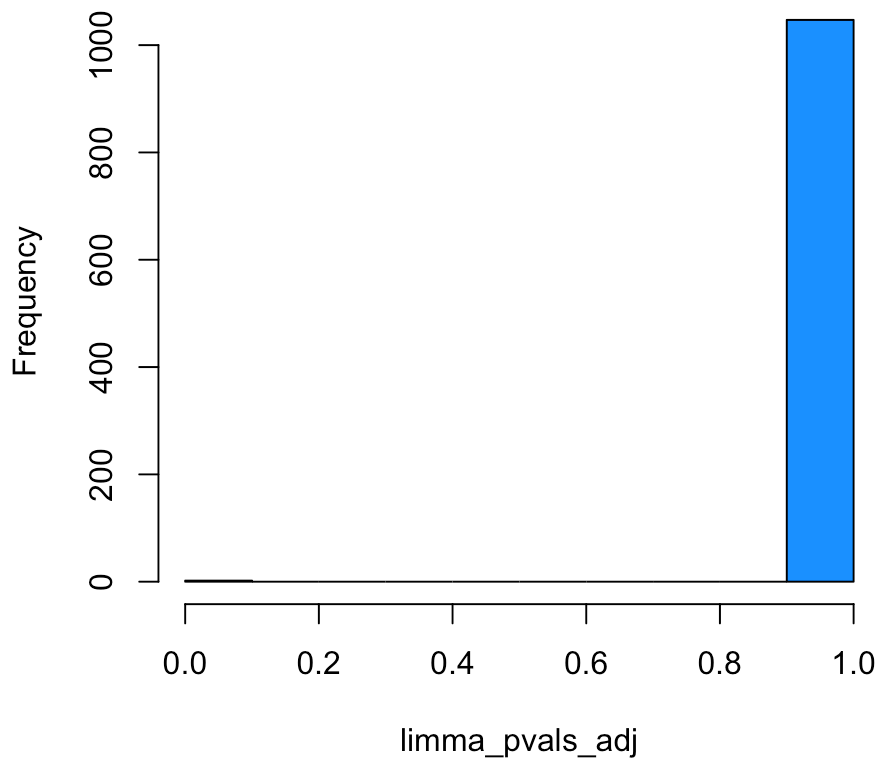
```
quantile(fp_bh)
```

```
##           0%          25%          50%          75%          100%  
## 0.1791152 0.8178625 0.8558812 0.9011016 0.9991666
```

Adjusted p-values from limma

```
limma_pvals_adj = topTable(ebayes_limma,number=dim(edata)[1])$adj.P.Val  
hist(limma_pvals_adj,col=2)
```

Histogram of limma_pvals_adj



```
quantile(limma_pvals_adj)
```

```
##           0%          25%          50%          75%         100%  
## 0.003270193 0.925591437 0.925591437 0.925591437 0.996066709
```

Direct q-values

```
qval_limma = qvalue(limma_pvals)  
summary(qval_limma)
```

```
##  
## Call:  
## qvalue(p = limma_pvals)  
##  
## pi0: 0.3930485  
##  
## Cumulative number of significant calls:  
##  
##           <1e-04 <0.001 <0.01 <0.025 <0.05 <0.1 <1  
## p-value      2      2      4      7     21    41 1049  
## q-value      0      0      2      2      2      2 1049  
## local FDR    0      0      1      2      2      2  2
```

```
qval$pi0
```

```
## [1] 1
```

q-values using edge

```
qval = qvalueObj(de_obj)
summary(qval)
```

```
##
## Call:
## qvalue(p = pval)
##
## pi0: 1
##
## Cumulative number of significant calls:
##
##           <1e-04 <0.001 <0.01 <0.025 <0.05 <0.1 <1
## p-value      0      0      1      2      8     20 1049
## q-value      0      0      0      0      0      0 1049
## local FDR    0      0      0      0      0      0  0
```

More information

Multiple testing is one of the most commonly used tools in statistical genomics.

- Statistical significance for genome-wide studies (<http://www.pnas.org/content/100/16/9440.full>) is a great place to start.
- The qvalue package vignette (<https://www.bioconductor.org/packages/release/bioc/vignettes/qvalue/inst/doc/qvalue.pdf>) is also informative.
- This is a nice review on How does multiple testing correction work? (<http://www.ncbi.nlm.nih.gov/pmc/articles/PMC2907892/>)

Session information

Here is the session information

```
devtools::session_info()
```

```

## setting value
## version R version 3.2.1 (2015-06-18)
## system x86_64, darwin10.8.0
## ui RStudio (0.99.447)
## language (EN)
## collate en_US.UTF-8
## tz America/New_York
##
## package * version date
## acepack 1.3-3.3 2014-11-24
## annotate 1.46.1 2015-07-11
## AnnotationDbi * 1.30.1 2015-04-26
## assertthat 0.1 2013-12-06
## BiasedUrn * 1.06.1 2013-12-29
## Biobase * 2.28.0 2015-04-17
## BiocGenerics * 0.14.0 2015-04-17
## BiocInstaller * 1.18.4 2015-07-22
## BiocParallel 1.2.20 2015-08-07
## biomaRt 2.24.0 2015-04-17
## Biostrings 2.36.3 2015-08-12
## bitops 1.0-6 2013-08-17
## bladderbatch * 1.6.0 2015-08-26
## broom * 0.3.7 2015-05-06
## caTools 1.17.1 2014-09-10
## cluster 2.0.3 2015-07-21
## colorspace 1.2-6 2015-03-11
## corpcor 1.6.8 2015-07-08
## curl 0.9.2 2015-08-08
## DBI * 0.3.1 2014-09-24
## dendextend * 1.1.0 2015-07-31
## DESeq2 * 1.8.1 2015-05-02
## devtools * 1.8.0 2015-05-09
## digest 0.6.8 2014-12-31
## dplyr * 0.4.3 2015-09-01
## edge * 2.1.0 2015-09-06
## evaluate 0.7.2 2015-08-13
## foreign 0.8-66 2015-08-19
## formatR 1.2 2015-04-21
## Formula * 1.2-1 2015-04-07
## futile.logger 1.4.1 2015-04-20
## futile.options 1.0.0 2010-04-06
## gdata 2.17.0 2015-07-04
## genefilter * 1.50.0 2015-04-17
## geneLenDataBase * 1.4.0 2015-09-06
## geneplotter 1.46.0 2015-04-17
## GenomeInfoDb * 1.4.2 2015-08-15
## GenomicAlignments 1.4.1 2015-04-24
## GenomicFeatures 1.20.2 2015-08-14
## GenomicRanges * 1.20.5 2015-06-09
## genstats * 0.1.02 2015-09-05
## ggplot2 * 1.0.1 2015-03-17
## git2r 0.11.0 2015-08-12

```

##	GO.db	3.1.2	2015-09-06
##	goseq	* 1.20.0	2015-04-17
##	gplots	* 2.17.0	2015-05-02
##	gridExtra	2.0.0	2015-07-14
##	gtable	0.1.2	2012-12-05
##	gtools	3.5.0	2015-05-29
##	highr	0.5	2015-04-21
##	HistData	* 0.7-5	2014-04-26
##	Hmisc	* 3.16-0	2015-04-30
##	htmltools	0.2.6	2014-09-08
##	httr	1.0.0	2015-06-25
##	IRanges	* 2.2.7	2015-08-09
##	KernSmooth	2.23-15	2015-06-29
##	knitr	* 1.11	2015-08-14
##	lambda.r	1.1.7	2015-03-20
##	lattice	* 0.20-33	2015-07-14
##	latticeExtra	0.6-26	2013-08-15
##	lazyeval	0.1.10	2015-01-02
##	limma	* 3.24.15	2015-08-06
##	lme4	1.1-9	2015-08-20
##	locfit	1.5-9.1	2013-04-20
##	magrittr	1.5	2014-11-22
##	MASS	* 7.3-43	2015-07-16
##	Matrix	* 1.2-2	2015-07-08
##	MatrixEQTL	* 2.1.1	2015-02-03
##	memoise	0.2.1	2014-04-22
##	mgcv	* 1.8-7	2015-07-23
##	minqa	1.2.4	2014-10-09
##	mnormt	1.5-3	2015-05-25
##	munsell	0.4.2	2013-07-11
##	nlme	* 3.1-122	2015-08-19
##	nloptr	1.0.4	2014-08-04
##	nnet	7.3-10	2015-06-29
##	org.Hs.eg.db	* 3.1.2	2015-07-17
##	plyr	1.8.3	2015-06-12
##	preprocessCore	* 1.30.0	2015-04-17
##	proto	0.3-10	2012-12-22
##	psych	1.5.6	2015-07-08
##	qvalue	* 2.0.0	2015-04-17
##	R6	2.1.1	2015-08-19
##	RColorBrewer	1.1-2	2014-12-07
##	Rcpp	* 0.12.0	2015-07-25
##	RcppArmadillo	* 0.5.400.2.0	2015-08-17
##	RCurl	1.95-4.7	2015-06-30
##	reshape2	1.4.1	2014-12-06
##	rmarkdown	0.7	2015-06-13
##	rpart	4.1-10	2015-06-29
##	Rsamtools	1.20.4	2015-06-01
##	RSkittleBrewer	* 1.1	2015-09-05
##	RSQLite	* 1.0.0	2014-10-25
##	rstudioapi	0.3.1	2015-04-07
##	rtracklayer	1.28.9	2015-08-19

##	rversions	1.0.2	2015-07-13
##	S4Vectors	* 0.6.5	2015-09-01
##	scales	0.3.0	2015-08-25
##	snm	1.16.0	2015-04-17
##	snpStats	* 1.18.0	2015-04-17
##	stringi	0.5-5	2015-06-29
##	stringr	1.0.0	2015-04-30
##	survival	* 2.38-3	2015-07-02
##	sva	* 3.14.0	2015-04-17
##	tidyr	0.2.0	2014-12-05
##	UsingR	* 2.0-5	2015-08-06
##	whisker	0.3-2	2013-04-28
##	XML	3.98-1.3	2015-06-30
##	xml2	0.1.2	2015-09-01
##	xtable	1.7-4	2014-09-12
##	XVector	0.8.0	2015-04-17
##	yaml	2.1.13	2014-06-12
##	zlibbioc	1.14.0	2015-04-17
##	source		
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## CRAN (R 3.2.1)
## CRAN (R 3.2.1)
## Bioconductor
## Github (alyssafranze/RSkittleBrewer@0a96a20)
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## Bioconductor
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It is also useful to compile the time the document was processed. This document was processed on: 2015-09-06.